

COOLING PUMP OPERATIONS MANUAL

Document ID: MAN-PUMP-001

Equipment Description

Cooling pumps circulate process water throughout the plant and are critical to safe operations.

Normal Operating Parameters

- Temperature: 40°C - 80°C
- Vibration: 2 - 5 mm/s
- Pressure: 90 - 130 bar
- Flow Rate: 180 - 300 m³/hr

Critical Thresholds

- Vibration > 7 mm/s
- Pressure < 80 bar
- Flow Rate < 150 m³/hr

Common Failure Modes

1. Bearing Wear
2. Mechanical Seal Failure
3. Cavitation
4. Impeller Damage
5. Motor Overheating

Troubleshooting Guide

Low Flow Rate

Possible Causes:

- Impeller damage
- Cavitation
- Suction blockage

Actions:

- Inspect impeller
- Verify suction conditions

High Vibration

Possible Causes:

- Bearing wear
- Misalignment

Actions:

- Inspect bearings
- Verify coupling alignment

Maintenance Recommendations

Daily:

- Verify pressure and flow

Weekly:

- Check vibration trends

Monthly:

- Bearing inspection

Failure Signature

Mechanical Seal Failure:

Indicators:

- Increased leakage
- Reduced pressure
- Temperature increase

Corrective Action:

Replace seal kit and verify shaft condition.